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Role: Computer Vision

ADVANCED 2D TRY-ON

Today, we focused on improving the **Virtual Try-On system** by enhancing garment fitting accuracy, realism, and body alignment techniques. The previous implementation mainly performed basic cloth overlay after body detection and pose estimation. In today's work, we improved the overlay mechanism by introducing **dynamic garment scaling**, better **shoulder alignment**, and integrating **transparent PNG clothing assets** for smoother blending with the user image.

We also worked on improving realism by applying **alpha blending techniques**, allowing garments to merge naturally with the body instead of appearing as a pasted image. Additionally, **cloth segmentation integration** was considered so that the system can understand clothing regions and determine accurate overlay positions automatically based on body posture and landmarks. Testing was performed using multiple images with different body positions to verify system performance.

Comparison Between Previous Task and Today's Task

Feature	Previous Task Output	Today's Improved Output
Virtual Try-On Type	Basic 2D overlay	Advanced 2D Virtual Try-On
Garment Fitting	Fixed cloth placement	Dynamic garment fitting
Scaling Method	Manual / Static resizing	Automatic dynamic scaling
Body Alignment	General body detection	Improved pose-based alignment
Shoulder Position	Less accurate	Accurate shoulder mapping
Clothing Images	Normal images	Transparent PNG garments
Blending Method	Direct overlay	Alpha blending integration
Cloth Processing	Simple placement	Cloth segmentation support

Feature	Previous Task Output	Today's Improved Output
Overlay Decision	Manual positioning	AI-based body understanding
Testing	Limited images	Multi-image testing
Output Quality	Moderate realism	Improved realistic visualization

Observation

Compared to the previous implementation, today's output produced **better cloth fitting**, improved garment positioning, and more realistic visualization. Dynamic scaling enabled automatic resizing according to body dimensions, while shoulder alignment reduced placement errors. The use of transparent PNG garments and alpha blending significantly improved the appearance of the virtual try-on output, making the clothing look more naturally integrated with the user image.

